Predict Pharma AI

Mediflow AI

DIMENSION DRIFTERS

Smart Drug Supply Chain & Shortage Prediction Platform

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Track/Domain : Healthcare / Supply Chain / AI

Problem Statement & Background

Understanding of the Problem

The Problem

Healthcare systems frequently face medicine shortages and inefficient supply management. Hospitals and pharmacies often rely on manual inventory systems, which leads to:

- Unexpected medicine shortages
- Overstocking and medicine expiration
- Poor coordination between hospitals
- Delayed response during emergencies

During crises like the COVID-19 pandemic, many hospitals ran out of essential medicines due to lack of predictive supply chain systems.

The Problem

Why It Matters

- Delayed treatments | Patient safety risks | Inefficient medicine distribution
- Stock shortages | Overstocking & expiry | Financial losses for hospitals
- Poor supply visibility | Slow emergency response | Healthcare inefficiencies

Who Affected

- Patients | Doctors & hospitals | Healthcare providers
- Pharmacies | Medical distributors | Supply chain operators
- Government health systems | Public health programs | Emergency response teams

Proposed Solution

Solution Overview

- A **Smart Drug Supply Chain** & Shortage Prediction Platform that provides real-time inventory visibility across hospitals and pharmacies.
- **Uses AI-based** demand forecasting to predict future medicine requirements.
- Generates early **shortage alerts** to prevent stockouts and redistribution between hospitals.

Key Features

- **Real-Time Tracking :** Hospitals can monitor stock levels through a centralized dashboard.
- **Predictive Analytics :** AI analyzes historical usage data to forecast upcoming demand.
- **Shortage Prevention :** Automated alerts notify hospitals before medicine stocks run out.
- **Smart Redistribution:** The system recommends transferring medicines

Problem Resolution

- **Real-Time Tracking:** Hospitals can monitor medicine through a dashboard .
- **Predictive Analytics:** AI analyzes historical usage data to forecast upcoming demand.
- **Shortage Prevention:** Automated alerts notify hospitals before medicine stocks run out.
- **Smart Redistribution:** The system recommends transferring medicines from surplus locations to shortage areas.

The Tech Stack & System Design

Key Components of Predict Pharma AI

Frontend

Streamlit or React

Backend

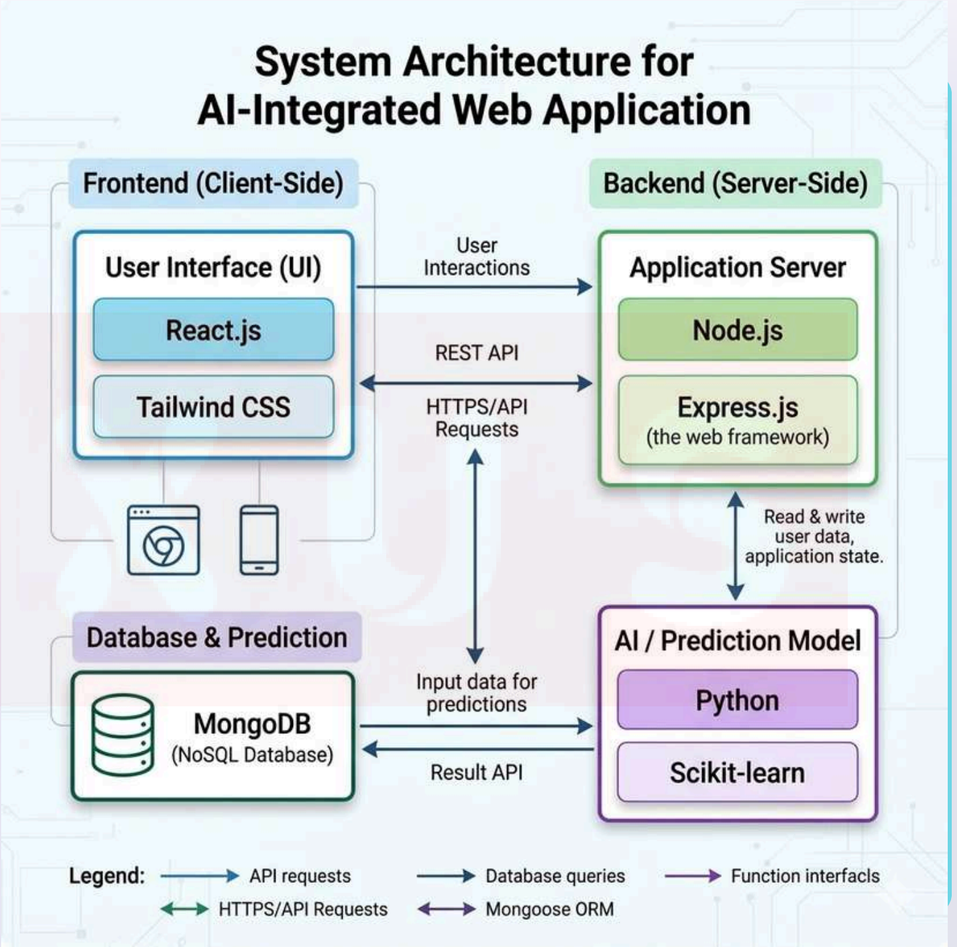
Python

Database

PostgreSQL & Redis or MongoDb

APIs & Services

Fast Api



Feasibility , Challenges & Mitigation

Addressing Key Hurdles in Implementation

1

Feasibility

The use of software-driven digital twins helps simulate real-world scenarios, allowing us to predict outcomes and optimize logistics while addressing data scarcity in medical supply chains.

2

Key Challenges

- Regulatory & Trust Barriers: Need high degree of transparency that is often missing in digital systems.
- Developing a Multi-Variable Matching Algorithm that balances distance, urgency, and medicine shelf-life (expiry) is a complex optimization problem.

3

Mitigation Strategy

- Use Point based Approach :
  High Points for medicine about to expire.
  High Points for hospitals with zero stock.
  Low Points for long dist.
- In hackthon we can't able to make use of scan barcode for expiry , we use other ways to demonstrate

Impact & Future Vision

Creating sustainable drug supply solutions

Expected Impact

- Improves patient care by ensuring continuous availability, Reduces medicine wastage, improves efficiency in hospital supply chains, Enables AI-driven demand prediction and data-driven healthcare logistics

Scalability

- Expand the platform to connect multiple hospitals, pharmacies, and distributors across regions.
- Integrate with government healthcare systems for large-scale medicine monitoring
- Use cloud infrastructure and advanced AI models to handle larger datasets and improve prediction accuracy.

Future Enhancements

- IoT Integration: Smart-crates with temperature and GPS sensors for cold-chain monitoring.
- Drone Delivery: Automated aerial redistribution for high-traffic areas.

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Team Roles & Responsibilities

Ankit Thakur

Role:

Full Stack development

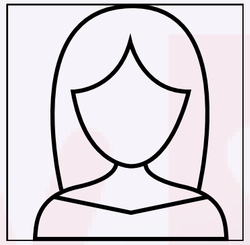
Responsibilities :
Help in front & backend , integrate them together

Saksham Thakur

Role:

Backend development

Responsibilities :
Responsible for most of the backend work



Sneha Rani

Role:

Frontend development

Responsibilities :
Responsible for most of the Frontend work

Yashmeen Paul

Role:

Research & Presentation

Responsibilities :
Responsible for research presentation & Ui deigning

Development Roadmap

Planning

Initial phase focused on defining objectives and project scope.

Development

Core MVP build with essential features and functionalities.

Integration

Mapping API integration for optimized logistics and real-time data.

Presentation

Final presentation at Frost Hack 2026, showcasing project outcomes.

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Thank You

Contact

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Repo Link : https://github.com/Ankitthakur-3500/INNOV-XUS-Hackathon-repo

Frost Hack 2026 | Cod-A-FestX

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